Forecasting Household Appliance Energy Demand

A Comparative Study of Benchmark, Classical, Machine-Learning, and Foundation-Model Forecasters

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Introduction

Accurately forecasting short-term household appliance energy demand is a practical problem for smart-home energy management, demand response, and grid-capacity planning. This report presents an end-to-end 24-hour-ahead forecasting study on the UCI Appliance Energy Prediction dataset (Candanedo, Feldheim, & Deramaix, 2017), which records energy use in a low-energy house in Belgium at 10-minute resolution over roughly 4.5 months, alongside indoor sensor readings and local weather observations.

This study aims more than just to benchmark one model, as it attempts to benchmark four classes of forecasting approach: (1) simple statistical benchmarks, (2) a classical time-series model (SARIMA), (3) a feature-based machine-learning model (XGBoost), and (4) a pretrained foundation model (Chronos) used in a zero-shot fashion. All four categories are assessed with the same 14 rolling-origin 24-hour test folds and error metrics (RMSE, MAE, MAPE) to eliminate the bias that may occur due to different measurements used to assess performance.

Data and Exploratory Analysis

The data was resampled from 10-minute resolution to hourly resolution (mean aggregation), and no missing data or gaps in the time series. Throughout, the target variable is Appliances, which is the average energy use of appliances per hour, in Wh.

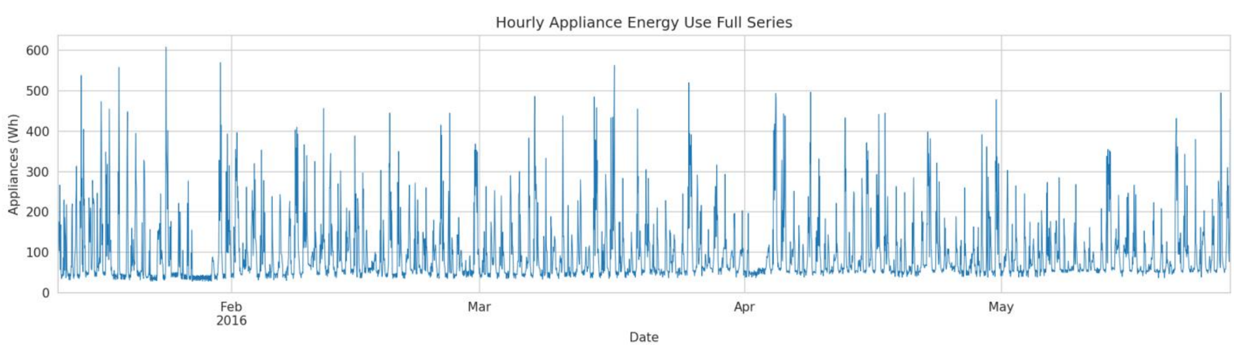

Figure 1: Full hourly Appliances series, Jan–May 2016

The strong daily time of day/week cycle identified by exploratory analysis (STL decomposition, hour-of-day and day-of-week averages, and ACF) showed (Figure 1) that the highest use was during the morning and evening peaks and the lowest use was overnight. A weaker but still present weekly cycle was also observed, which is a difference between the routines of the weekdays and the weekends.

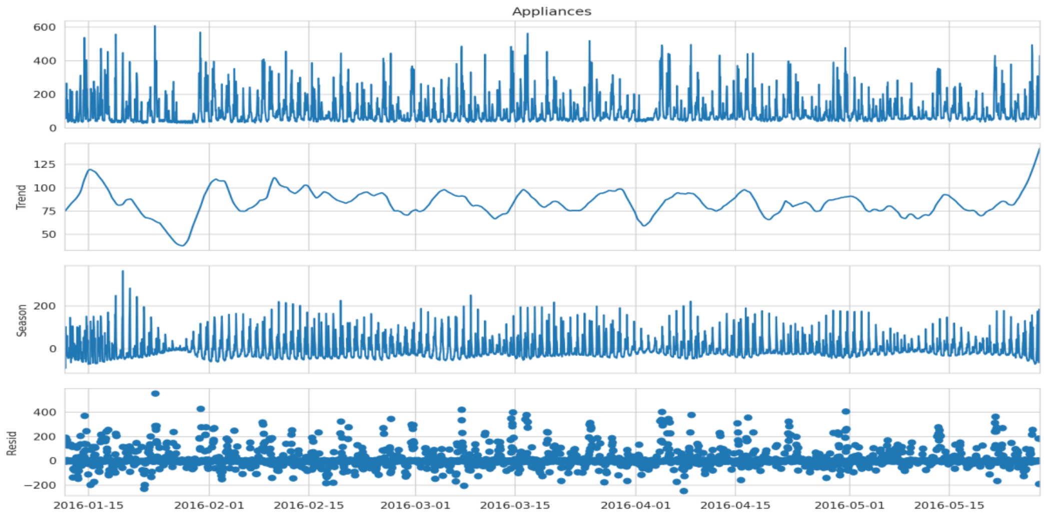

Figure 2: STL decomposition (trend, daily seasonal, residual) of the Appliances series

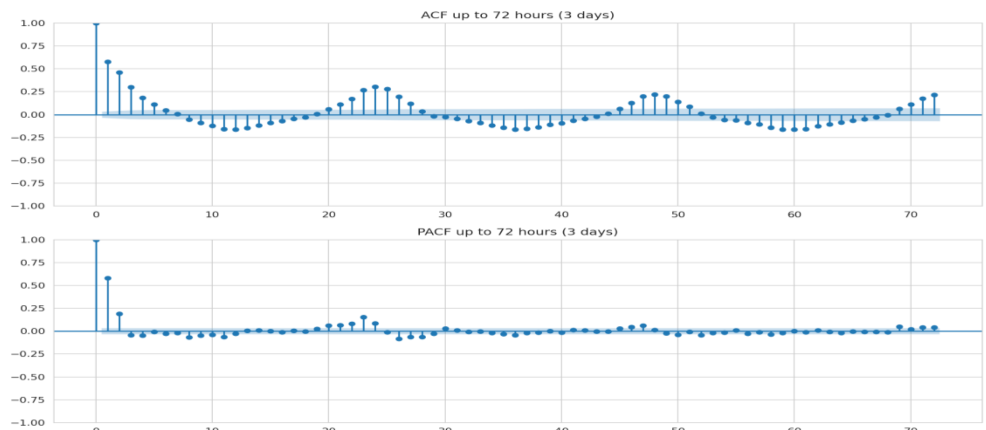

Figure 3: ACF and PACF of Appliances up to 72 hours, showing repeating spikes at multiples of 24 hours

The series was approximately stationary in levels, as indicated by the ADF and KPSS tests,Neither log transformation nor seasonal differencing was used in the final forecasting pipeline both were examined during exploratory/stationarity analysis. However, the strong 24-hour cycle observed in the EDA motivated the use of a seasonal SARIMA specification with a seasonal period of s = 24.

Methodology

Forecasting task definition

The task is formulated as: predict Appliances at a 24-hour time horizon, hourly time step, from any time in the past. The last 14 days (336 hours) were reserved for a test period and divided into 14 non-overlapping 24-hour rolling-origin folds, instead of using a one-time training/test split, to prevent the risk of a single chance of luck or bad luck in a single window and, therefore, allow each model its own opportunity to perform and report the result as mean ± std.

Benchmarks

Five simple benchmarks were added as reference base: Mean (forecast = historical average); Naive (last observed value repeated); Daily Seasonal Naive (repeats the value from 24 hours prior); Weekly Seasonal Naive (repeats the value from 168 hours / 7 days prior); Drift (naive plus a linear trend extrapolated across the training history).

SARIMA

Based on the exploratory analysis, the seasonal period was fixed at 24 hours, and an AIC grid search was performed over $p \in [0,6]$, $d \in [0,2]$, and $q \in [0,6]$, resulting in 147 combinations. The final selected order was SARIMA (2,1,6)×(1,0,1,24).To avoid recalculating the expensive seasonal maximum-likelihood optimization 14 times, a single reference model was fitted on the full training set. During walk-forward evaluation, the model state was updated with the newly observed values for each fold without refitting or re-estimating the model parameters.

XGBoost (feature-based ML model)

A direct multi-horizon design was employed, since training one model to make one step prediction at a time (recursively) would accumulate errors over 24 steps. A single XGBoost regressor was trained on a long-format dataset, where each row contains an origin time t_0, a horizon $h \in [1,24]$, and a target value v at t_0+h, with the horizon h itself added as a feature. The weather/sensor readings were current (t_0 only), and the calendar features included hour, day-of-week, month, weekend flag, and cyclical sine/cosine encodings, where the latter were computed over the given window of time. The weather/sensor readings were current (t_0 only), and the calendar features were hour, day-of-week, month, weekend flag, and cyclical sine/cosine encodings (computed over the specified window of time). More critically, no future weather values were used as features, as this would not be true in a real deployment where such values would not be truly available at forecast time (see Section 5.5).

Chronos (foundation model)

Chronos-T5-small (Ansari et al., 2024) is a transformer that was pre-trained on a large and diverse dataset of time series, and was used in the zero-shot setting without any training or fine-tuning on the Chronos-T5-small test set. The model was provided with the 512 hours of previous Appliances data as context for each fold and required to generate a probabilistic prediction for the next 24 hours of data; the median of 20 sampled forecasts was taken as the point forecast.

Results

Overall comparison

Table 1 reports the mean metrics across all 14 folds for all 8 models, ranked by RMSE.

Table 1: Final model comparison (mean across 14 rolling-origin folds)

Rank	Model	RMSE (Wh)	MAE (Wh)	MAPE (%)
1	SARIMA (2,1,6)×(1,0,1,24)	56.70	37.59	34.85
2	XGBoost (direct multi-horizon)	61.02	44.04	43.73
3	Chronos-T5-small (zero-shot)	61.71	36.02	23.26
4	Mean	66.91	50.26	53.70
5	Weekly Seasonal Naive	69.23	43.46	37.35
6	Daily Seasonal Naive	76.30	48.31	43.33
7	Naive	97.77	85.55	112.91
8	Drift	97.98	85.80	113.31

Benchmarks

Among the five benchmarks, Mean achieved the best RMSE (66.91 Wh), while Weekly Seasonal Naive achieved the best MAE (43.46 Wh) and MAPE (37.35%), well ahead of Daily Seasonal Naive, with Naive and Drift far weakest (RMSE ≈ 98). This split is informative: Mean avoids large misses by regressing to a stable average, while Weekly Seasonal Naive tracks typical usage more closely by exploiting real weekly-cycle structure beyond the daily cycle. Naive and Drift performing so poorly confirms the series is far from a random walk.

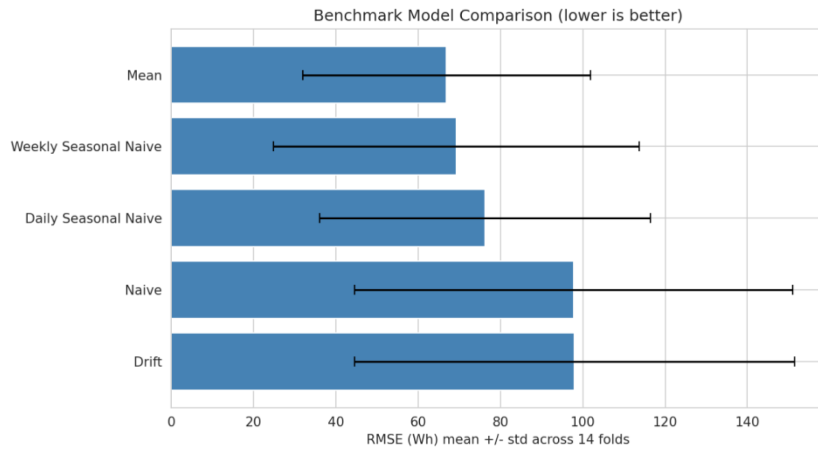

Figure 4: Benchmark comparison by mean RMSE across 14 folds

SARIMA

The seasonal AIC grid search selected SARIMA (2,1,6)×(1,0,1,24) with an AIC of 5101.43 on the grid-search subset, indicating that this specification provided the best fit among the models evaluated in the seasonal grid search. The model that performed best was SARIMA, with an RMSE of 56.70 Wh, which was 18% lower than the Weekly Seasonal Naive. The residual ACF show (Figure 5) no obvious significant spikes and the reported Ljung-Box lag-1 test is non-significant. However, the residuals seem to be visually heavy-tailed, suggesting that the model captures the usual hours well but fails to capture larger usage bursts.

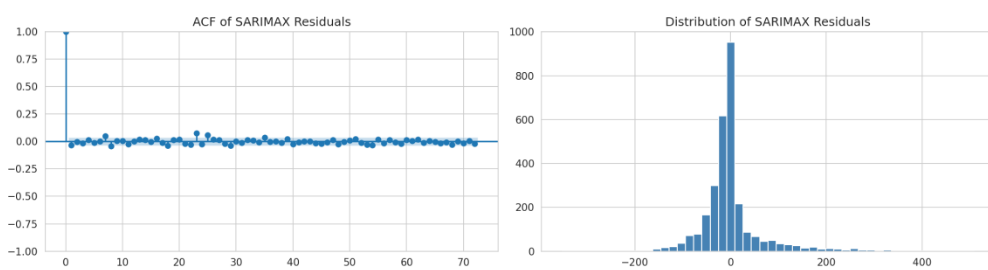

Figure 5: ACF and distribution of SARIMAX residuals, no significant remaining autocorrelation, but heavy-tailed residuals

Figure 6 plots this directly on a held-out fold, which contains a sharp spike around ~223 Wh, and SARIMA does a very good job of following the low, stable overnight time period, with its 95% CI definitely and noticeably containing the true value around the time of the two spikes, even though it misses the value entirely at both times.

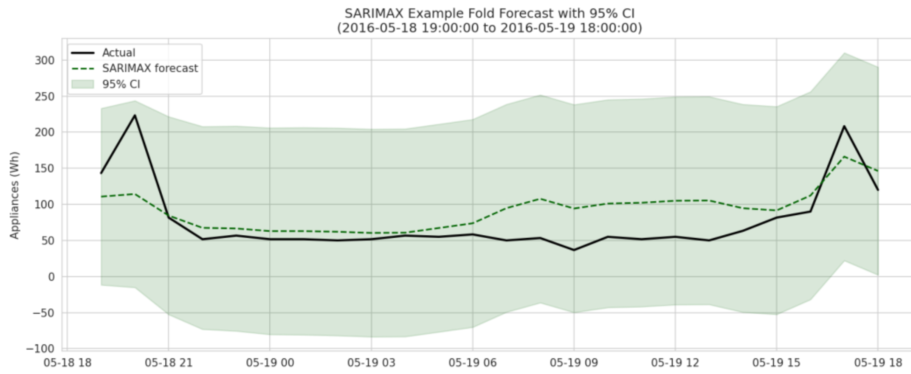

Figure 6: SARIMAX 24-hour forecast vs. actual data collected after the forecast origin, with 95% CI

XGBoost

XGBoost achieved lower average error than Mean, Naive, Drift, and Daily Seasonal Naive, demonstrating that the engineered features are indeed predictive, but did not outperform SARIMA on any measure, and only beat Weekly Seasonal Naive on RMSE. A very clear pattern emerged for feature importance (Figure 7): calendar/time-of-day features, such as hour, hour_sin/cos, day-of-week, and month, top the ranking, ahead of all raw lag features, and the first non-calendar feature to appear in the ranking is the 168-hour rolling mean. The six raw lag features (lag_1, lag_2, lag_3, lag_24, lag_48, and lag_168) rank toward the lower part of the feature-importance plot, while calendar features dominate the highest ranks.

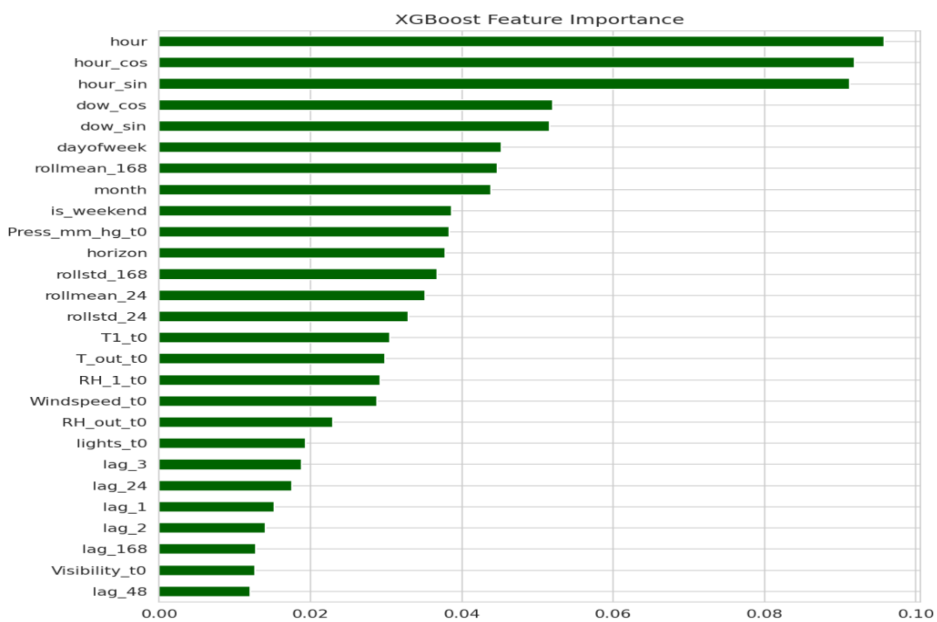

Figure 7: XGBoost feature importance calendar features dominate over raw lags

This is because of the direct multi-horizon design: The same features must be used across all horizons 1 to 24, so the features 'what time of day is the forecast' and 'what time of week is the forecast' are informative at all horizons. In contrast, the feature 'lag_1' is highly informative at horizon 1 and uninformative at horizon 24. This is also the reason XGBoost does not outperform SARIMA: SARIMA takes advantage of short-lag autocorrelation at every step as part of its state-space recursion, whereas XGBoost's design does not use that correlation at every step, instead averaging across all 24 horizons.

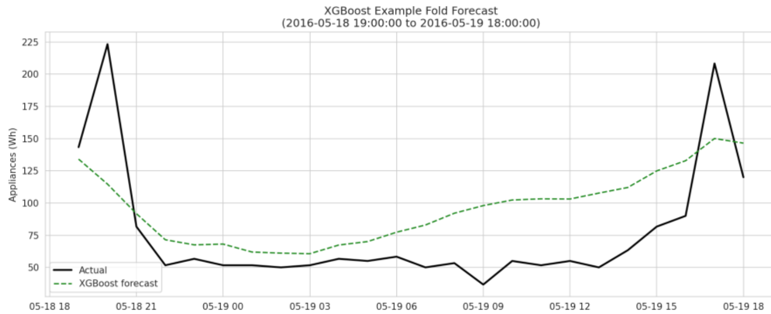

Figure 8: XGBoost 24-hour forecast vs. actual data on the same fold

Chronos (zero-shot foundation model)

Despite no training on this dataset whatsoever, Chronos achieved the lowest MAE (36.02 Wh) and MAPE (23.26%) of all eight models, and was close but slightly behind SARIMA on RMSE (61.71 Wh). Its relatively high RMSE standard deviation value (42.97 Wh, which is the highest among all the models) reveals the explanation: Chronos is very accurate on typical hours, and the occasional larger miss will raise the RMSE value, but will not equally raise MAE or MAPE.

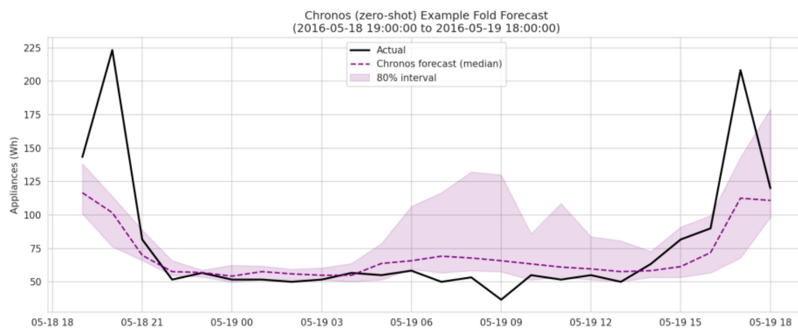

Figure 9: Chronos zero-shot 24-hour forecast vs. actual data, with 80% predictive interval

On the same fold shown in Figures 6, 8, and 9, Chronos tracks the low nightly baseline in addition to SARIMA and its 80% interval, which, although narrower than SARIMA's 95% CI, covers most of the true trend, only being significantly exceeded during the two steep spikes, which are not fully captured by any model in this study.

Cross-model comparison

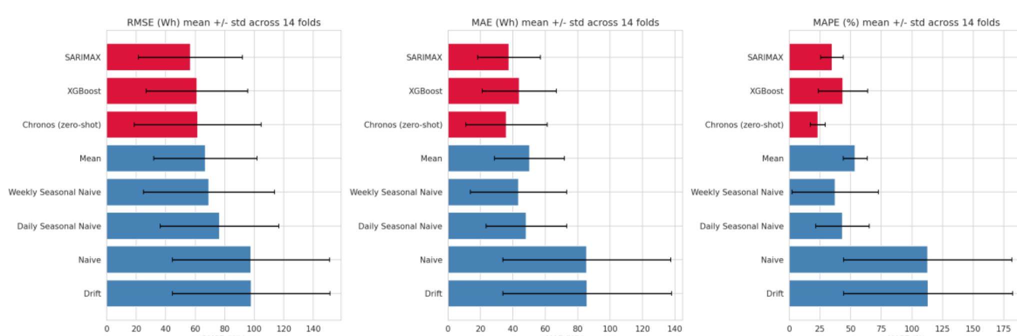

Figure 10: RMSE, MAE, and MAPE compared across all 8 models (red = fitted/pretrained models, blue = simple benchmarks).

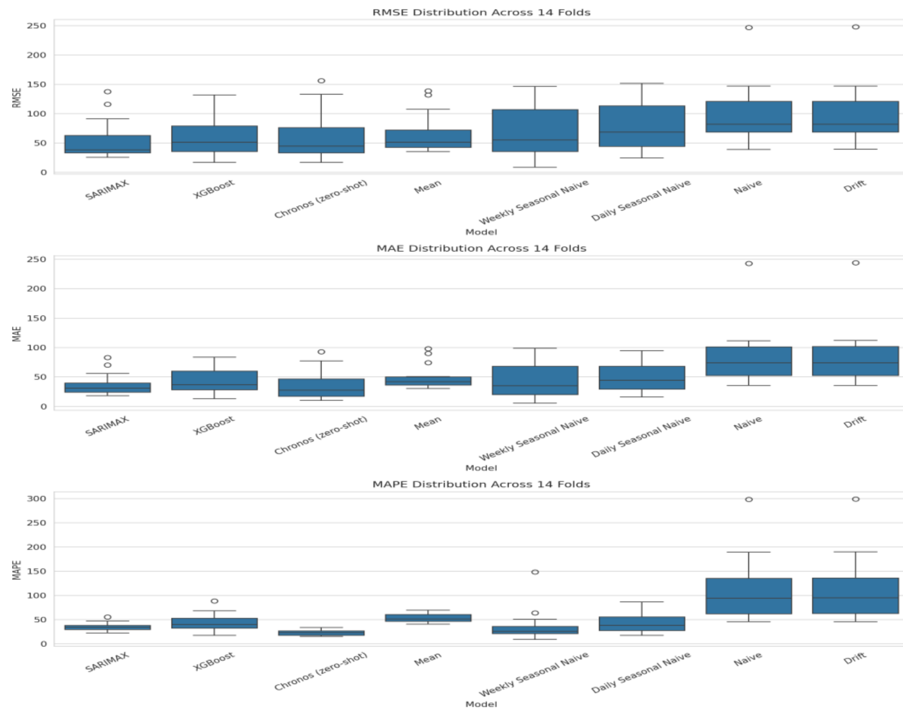

Figure 11: Per-fold error distributions across all 8 models showing consistency, not just average accuracy

Figure 11 adds an important nuance beyond the mean metrics in Table 1: SARIMA's box is narrow and consistent, explaining its RMSE win, while Chronos's MAPE box is both the lowest and one of the tightest of any model the most reliable typical-case performer fold after fold, not just on average.

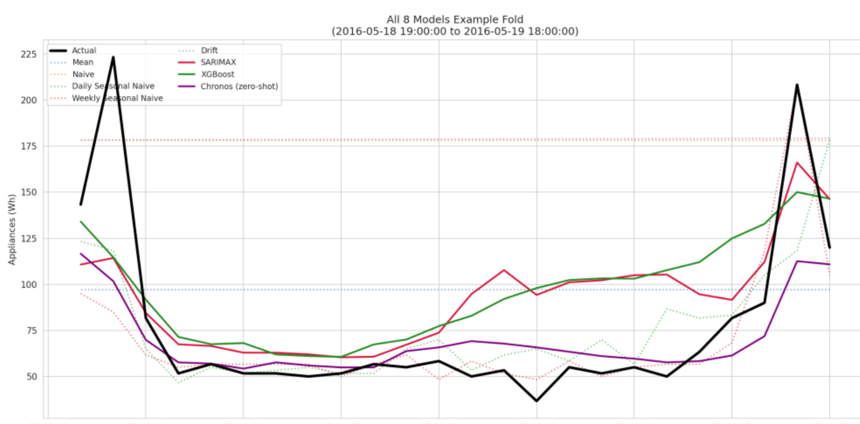

Figure 12: All 8 models' 24-hour forecasts overlaid against the actual held-out data for one representative fold

Figure 12 encapsulates the main results of this study visually: All models fail to capture the two sharp peaks of this fold, though SARIMA and Weekly Seasonal Naive respond quickly to the increase in usage, whereas XGBoost is less responsive to the rise in the second peak and Chronos is also slow to catch the increase in the usage in the second peak, though smooth and well-calibrated during the flat periods.

Discussion and Critical Analysis

Model-level discussion

The results show a true trade-off between the models, and not just one 'best' model: although SARIMA has the most efficient, well-documented classical structure and completely disregards the exogenous information, it performs well in situations where the model fits the data; XGBoost, the model with the most features, is not as effective as its competitors in the cases where it accurately predicts the true value, but is far worse in cases where it misses the true value by a large margin; and Chronos, which has no training on this data, performs best on a typical case basis, at the expense of much larger misses when the model doesn't fit the data. All four categories of models in Figures 6, 8, 9, and 12 perform well on the stable periods, and the four models share a common characteristic: they do not fully predict the usage peaks that occur during the series, as is shown by the heavy-tailed residuals in the SARIMA model (Section 4.3) and as is expected of a series of near-random usage bursts.

Answers to the Six Analysis Questions

Q1. Which benchmark model is strongest, and what does this tell you about the structure of appliance energy use? Among the five benchmarks, Mean achieved the best RMSE (66.91 Wh), while Weekly Seasonal Naive achieved the best MAE (43.46 Wh) and MAPE (37.35%), well ahead of Daily Seasonal Naive, with Naive and Drift far weakest (RMSE ≈ 98). This split is informative: Mean avoids large misses by regressing to a stable average, while Weekly Seasonal Naive tracks typical usage more closely by exploiting real weekly-cycle structure beyond the daily cycle. Naive and Drift performing so poorly confirms the series is far from a random walk.

Q2. Does SARIMAX improve on the strongest seasonal benchmark? Are daily seasonality, autocorrelation, and exogenous variables adequately captured? Yes,The performance of SARIMA is superior to Weekly Seasonal Naive across all three metrics: 18% reduction in RMSE, 14% reduction in MAE, and 7% reduction in MAPE. The appearance of an explicit daily seasonal term, in addition to the ARMA structure alone, is an advantage of the model selected, SARIMA(2,1,6)×(1,0,1,24) (AIC 5101.43 on the grid-search subset), and short-lag autocorrelation is well modelled. This SARIMA model, however, is purely univariate, with a noted limitation being that exogenous variables are not captured at all.

Q3. Does the feature-based model improve with lag, rolling, calendar, and sensor features, and which feature groups are most useful? Partially XGBoost significantly outperforms Naive, Drift, and Daily Seasonal Naive but not SARIMA. The features in general are mostly calendar/time-of-day features, much more important than all the raw lag features, with the 168h rolling mean being the first non-calendar feature to show up in the top ranks (Figure 7). This is the direct multi-horizon design: One from horizon 1 is informative from horizon 24, one from horizon 24 is informative from horizon 24, and so on.

Q4. Does the foundation model outperform the simpler, SARIMA, and feature-based models and is any improvement large enough to justify its complexity? Yes, on typical-case accuracy: Chronos' MAE is 36.02 Wh, which is by far the lowest, and its MAPE is 23.26%, which is significantly lower than the next-lowest, 34.85%, for SARIMA; its RMSE, 61.71 Wh, is slightly higher than the lowest (56.70 Wh) of SARIMAX. It's the same data, no feature engineering, no tuning, and a significant improvement, so it is worth the added complexity, especially since it is at inference time, not at development time.

Q5. Which variables are genuinely known at the forecast origin? Would using future weather from the test set make this a true or a conditional forecast? Only calendar features (hour, day-of-week, month, and their cyclical encodings) are genuinely known in advance for any future timestamp. Weather/sensor readings are known only up to t_0. Using the test set's actual future weather values as inputs rather than only current (t_0) readings, In this study, only weather/sensor values available at the forecast origin were used. If actual future weather values from the test set were instead supplied to the model, the task would become a conditional forecast rather than a true autonomous forecast.

Q6. Based on accuracy, interpretability, uncertainty, computational cost, and ease of deployment, which model would you recommend for practical smart-home energy forecasting? See Table 2 below. Recommendation: Chronos (zero-shot) despite its relatively low interpretability.Where reliability or auditability is more important, and the worst-case scenario is more important (e.g., grid capacity planning), then SARIMA is the better option as it has the best RMSE and clear structure and can be diagnosed. XGBoost is the least defensible of the three fitted models here: The three most expensive models to engineer, and it doesn't clearly outperform either of the other two.

Table 2: Model trade-offs for practical deployment (Q6)

Criterion	SARIMAX	XGBoost	Chronos (zero-shot)
Accuracy	Best RMSE (56.70)	Middling on all metrics	Best MAE/MAPE, competitive RMSE
Interpretability	High explicit order, diagnostics	Medium feature importances only	Low black-box transformer
Uncertainty	Principled CIs (state-space)	None natively	Sample based 80% predictive interval
Deployment ease	Cold-start issue (needs history)	Cold-start issue + pipeline upkeep	No cold-start works immediately

Future Improvements

1. Add exogenous regressors to the SARIMA (true SARIMA model) using current weather/sensor readings, and test whether classical structure + covariates can bridge the gap with Chronos in terms of MAPE.
2. Train per-horizon specialist models rather than one direct multi-horizon XGBoost to see if the short-lag signal that is lost in the current design (Section 4.4) is restored, and tune hyperparameters using cross-validated search.
3. Test larger variants of Chronos or other foundation models like TimesFM (Das et al., 2024) and calculate the accuracy/inference-cost difference between CPU and GPU.
4. Incorporate conformal prediction in XGBoost for native uncertainty intervals and expand with longer historical record, as zero-shot foundation models are likely to have a relative advantage precisely when task-specific data is limited.

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